

A 3D Interactive Framework for Spatially Coincident Geodata: A Design Study of Origin-Destination Data

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Abstract—Origin-Destination (OD) analysis is fundamental for Intelligent Transportation Systems, but visualizing flow data for Areas of Interest (AOIs) can cause a “spatial coincidence” conflict, where inflow and outflow magnitudes occupy the same geographic coordinates at the same time. This paper introduces “Pillars,” an interactive 3D visualization framework designed to reduce overdraw problems and facilitate comparison of traffic volumes. Using a problem-driven design study approach, six visual encodings were systematically tested. Results indicate that linear 3D encoding combined with a constrained interaction model greatly reduces perceptual overload compared to traditional 2D aggregation or edge bundling. The resulting system offers a scalable, perceptually grounded approach for modern urban analytics. A web-based implementation using MapLibre and deck.gl is provided to demonstrate the framework’s real-time performance [Live Demo].

Index Terms—spatial coincidence; geospatial visualization; origin-destination flows; information visualization; human-computer interaction

I. INTRODUCTION

Modern Intelligent Transportation Systems (ITS) now draw on a wide range of data sources from government records, sensor networks, camera streams, and various ITS logs, and the volume continues to grow as cities deploy more digital infrastructure. Zheng et al. [1] describe this model as “urban computing”, where the primary challenge shifts from gathering data to interpretation. The size and variety of traffic records can make them difficult to analyze. This difficulty is evident in pioneer cities. For example, Hangzhou, as the initial implementation site for the “City Brain” initiative, features

advanced digital infrastructure [2]. The technical integration of these multi-source datasets has significantly improved. However, practitioners frequently encounter perceptual limitations rather than computational ones. Traditional mapping techniques struggle to convey this density effectively [3].

Origin-Destination (OD) analysis captures the “tidal” pulse of urban commuting, enabling planners to diagnose congestion’s root causes rather than just symptoms [4]. Effective governance demands simultaneously comparing inflow and outflow volumes to assess local traffic burdens. However, mapping these dual vectors to a single location create a topographic conflict. This conflict, termed spatial coincidence, arises when inflow and outflow magnitudes map to identical geographic coordinates. An OD node acts like a visual singular point on a 2D choropleth or symbol map because the Origin and Destination attributes are at the same (x,y) coordinates. This dimensional collapse means that no amount of 2D jittering or transparency can completely resolve the occlusion without changing the underlying geographic truth.

A. Contributions

To resolve this topographic conflict, I present “Pillars,” a three-dimensional interactive framework developed through a problem-driven design study [5] utilizing 400,000 mobility records from the Dios Platform. While 3D visualization is often criticized for perspective distortion [6], recent empirical studies suggest that 3D bars support accurate quantitative comparisons compared to area or volume encodings [7].

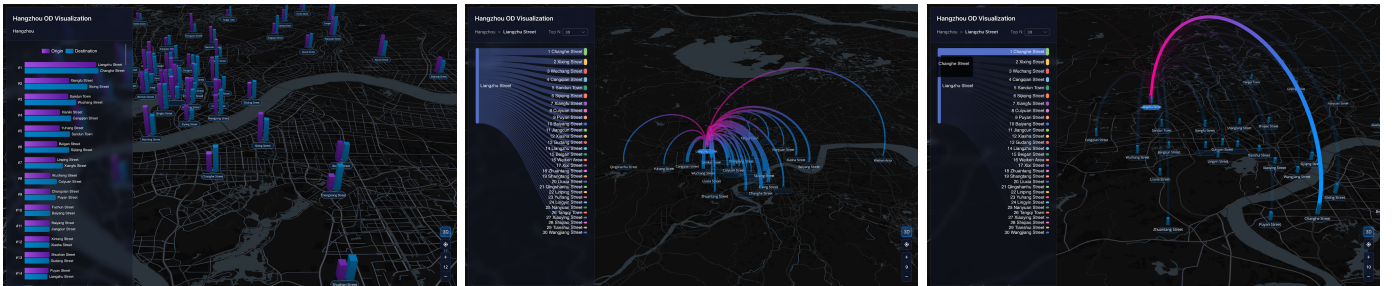


Fig. 1. The Pillars Interface. (a) Initial State. The left side displays 2D ranking bars, while the right side shows 3D pillars that represent OD totals on the map. (b) Focus State. Clicking a pillar/bar transitions the view to a detailed Sankey diagram and generates 3D flow arcs. (c) Highlight State. Clicking an arc/link from the Sankey, the arc and the link will be highlighted. A live interactive demo is available at [https://od-visualization-32uf.vercel.app/]. (Note: Visualization depicts synthetic data for privacy protection.)

Building on this finding, this paper argues that vertical extrusion uniquely mitigates over-plotting in dense urban networks without sacrificing geospatial integrity.

My contributions are twofold:

- **Theoretical & Design Contribution:** I introduce the “Pillars” framework, a novel visual encoding designed to resolve the topographic conflict of spatial coincidence.
- **Technical Contribution:** I developed a validated rendering architecture combining MapLibre and deck.gl. Through custom texture-atlas optimizations and instanced rendering, the system achieves real-time performance for city-scale datasets, bridging the gap between theoretical design and industrial scalability [8].

II. RELATED WORK

A. Visualization of OD Data:

Movement is a spatiotemporal phenomenon involving changes in an object’s location and attributes. This is often represented as Origin-Destination (OD) movement, which has distinct beginning and ending points. OD data typically consists of three parts: a spatial component, a temporal component, and an attribute components. Although flow maps are widely used to encode these elements, their visual complexity frequently causes serious problems. Visual complexity refers to the level of intricacy that impedes readability. The number of possible connections rises quadratically with dataset size, frequently collapsing into hard-to-understand “spaghetti-like” patterns.

Experts in the field have proposed several ways to reduce this complexity. As illustrated in Figure 2, numerous 2D methods prioritize spatial reorganization and optimization. There are three main types of these: location aggregation, surface generation, and edge rerouting. **Location-aggregation** methods aggregate adjacent areas to reduce flow volume but obscure local patterns and create the well-known Modifiable Areal Unit Problem (MAUP), where arbitrary changes in aggregation boundaries can significantly distort the statistical properties of the underlying flow data [9]. **Surface-based** methods show how things move between nearby areas. While this reduces clutter, it obscures specific origin-destination pairings [10]. Finally, **edge bundling** techniques improve the

layout by changing the shapes of flow symbols. This improves legibility for smaller datasets, but bundled paths make it hard to find specific OD connections without making them stand out (Figure 2c). As a result, these methods often make data less accurate and don’t keep exact OD relationships.

Subsequent studies extended these approaches into 3D to leverage the vertical axis. 3D visualizations generally have fewer intersections or overlaps than 2D visualizations. Yang et al. [11] showed that lifting arcs above the map surface reduces visual congestion and improves task performance. Additionally, the third dimension can be used to encode other components, such as time or attributes. Gu et al. [12] highlight that 3D extrusions can efficiently decouple overlapping flows. Furthermore, Zuo et al. [13] demonstrated the efficacy of 3D space-time cubes (STC) in immersive environments. By mapping time to the vertical axis, their approach allows users to identify localized temporal hotspots that remain ambiguous in planar views.

Most previous research focuses on global network topology to clarify static, city-wide trends. However, this macro-level optimization is insufficient for operational requirements: specifically, the ability to compare inflow and outflow magnitudes for 193 discrete AOIs. As illustrated in Figure 3, edge-based techniques like bundling fail to resolve the visual ambiguity at the anchor points. Even if the paths are organized, the node-level conflict remains. Consequently, this study shifts focus from optimizing network paths to resolving value coincidence at the nodes themselves.

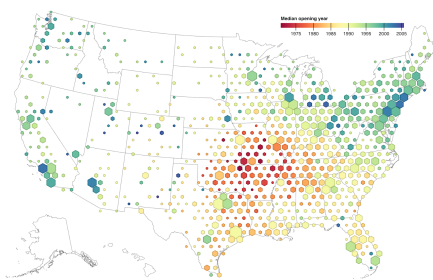
B. The Perceptual Science of Coincidence

Handling coincident data is a foundational problem in cartography. Standard solutions include aggregation (e.g., kernel density estimation), which sacrifices individual variation for overview, and small multiples, which separate attributes into different panels. While small multiples avoid occlusion, they demand significant screen space and force the user to perform mental integration across disparate views—a task known to incur high cognitive cost [14].

Alternative approaches involve composite glyphs [15] or displacement techniques (often termed ‘spiderfication’) that

Hexbin map

This map shows approximately 3,000 locations of Walmart stores. The hexagon area represents the number of stores in the vicinity, while the color represents the median age of these stores. Older stores are red, and newer stores are blue.



wind map

Nov. 9, 2025
3:44 pm EST
(live & weather forecast)

top speed: 27.9 mph
average: 8.8 mph

1 mph
2 mph
4 mph
6 mph
10 mph
12 mph
20 mph

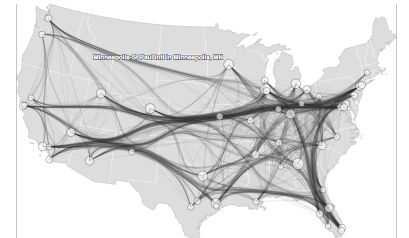
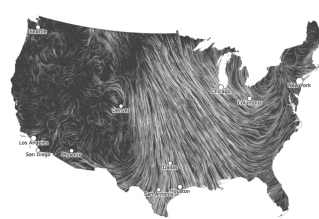


Fig. 2. Visual taxonomy of traditional 2D clutter-reduction techniques. (a) Location Aggregation: An example of hexagonal binning where individual data points are aggregated into discrete cells (image adapted from D3.js). (b) Surface Generation: A continuous vector field visualization (image adapted from Hint.fm/wind). (c) Edge Bundling: A force-directed layout that groups trajectories into thick bundles (image adapted from S. Engle).

push overlapping points outward. However, Korpi et al. [16] note that displacing symbols disrupts the user’s mental map of the geography. If a symbol representing the “Central Business District” is shifted into the adjacent “Residential District” to avoid overlap, the visualization actively misleads the user. The Pillars framework is predicated on the principle of spatial stability: markers must remain anchored to their semantic coordinates.

III. METHODOLOGY

This research was conducted as a specialized sub-project within the broader ‘Green Wave 1.3’ initiative—a major iteration of the company’s core product. While the 35-person interdisciplinary team, which comprised backend and frontend engineers, UI designers, and PMs, focused on the platform’s infrastructure and data pipelines, this specific design study targeted the visualization challenges of the Origin-Destination (OD) analysis module.

Within this framework, I served as the sole visualization lead, responsible for the conceptual design, the development of 3D rendering algorithms (via MapLibre/deck.gl), and the implementation of the final interactive interface.

A. Dataset

The primary dataset was sourced from the Dios Platform and covers the Hangzhou metropolitan area, China.

- **Volume:** Approximately 400,000 anonymized trip records.
- **Temporal Filter:** To isolate structural commuting patterns, weekend trips were removed. Analysis focused on weekday morning peaks (07:20–10:00) and evening peaks (16:00–19:30).
- **Spatial Aggregation:** Trip endpoints were mapped to 193 AOIs defined by official administrative block boundaries.

This aggregation strategy is central to the design problem. By collapsing tens of thousands of GPS points into 193 centroids,

the preprocessing step effectively induces spatial coincidence. This practice reflects the operational reality of urban planners, who manage districts, not coordinates.

B. Note on Data Privacy and Synthetic Representation

While the design study, performance benchmarking, and heuristic evaluations were conducted using the proprietary dataset (400,000 records) described above, strict confidentiality agreements prohibit the public release of specific user trajectories. Consequently, all figures in this manuscript and the accompanying live demonstration utilize a synthetic dataset. This synthetic data was algorithmically generated to replicate the topological density, distribution characteristics, and N:N complexity of the original logs, ensuring that the visual challenges represented remain faithful to the real-world problem.

C. Research Questions

To address the challenges identified in the related work, this study investigates three primary research questions:

- **RQ1:** How can multiple scalar values at the same coordinate be perceptually separated without distorting their geographic meaning?
- **RQ2:** Which visual encoding yields the highest accuracy for quantitative comparison (inflow vs. outflow)?
- **RQ3:** How can interaction manage the visual complexity of dense N:N flow networks?

IV. DESIGN EXPLORATION AND COMPARATIVE ANALYSIS

The development of the pillars framework was not linear; it was the result of a rigorous elimination process involving distinct design iterations. As illustrated in Figures 4 & 5, we evaluated six visual encodings against heuristics of perceptual accuracy, spatial stability, and cognitive load.

A. 2D Circles with Dynamic Offset

To establish a baseline, the first prototype adopted a standard proportional-symbol design common in commercial dashboards. Two circles were placed adjacent to the AOI centroid: the left circle represented Origin flow, and the right represented Destination flow. This design worked only when many AOIs were hidden around zoom level 15 or 16. As seen in the “Overview Map” column of Figure 4, once the view expanded to show larger districts or all 193 AOIs, like zoom level 10 (approx. 1:150,000 scale), the circles began to collide with nearby symbols, and the map quickly became cluttered. Adjusting the offset did not solve the problem.

To mitigate overlap, a dynamic recentering algorithm was implemented that recalculated the distance from the circles to the center of AOIs at each zoom level. However, this introduced disruptions to spatial continuity. Studies on mental-map preservation [17] indicate that users exhibit significant sensitivity to alterations in spatial reference. According to recent studies on visual stability, constant spatial shifts disrupt the user’s “mental map”, significantly increasing the cognitive load required to reorient familiar regions [18].

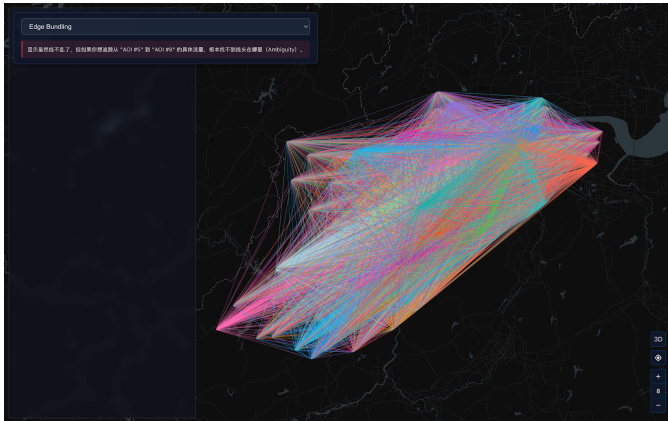


Fig. 3. Visualization of raw many-to-many (N:N) flows using the study’s dataset (Hangzhou). Even with transparency applied, the massive occlusion of overlapping edges creates a “spaghetti” effect. This dense connectivity effectively masks the underlying geographic context and makes it challenging to discern the magnitude of flow at any specific origin or destination node, confirming the limitations of purely edge-based 2D techniques.

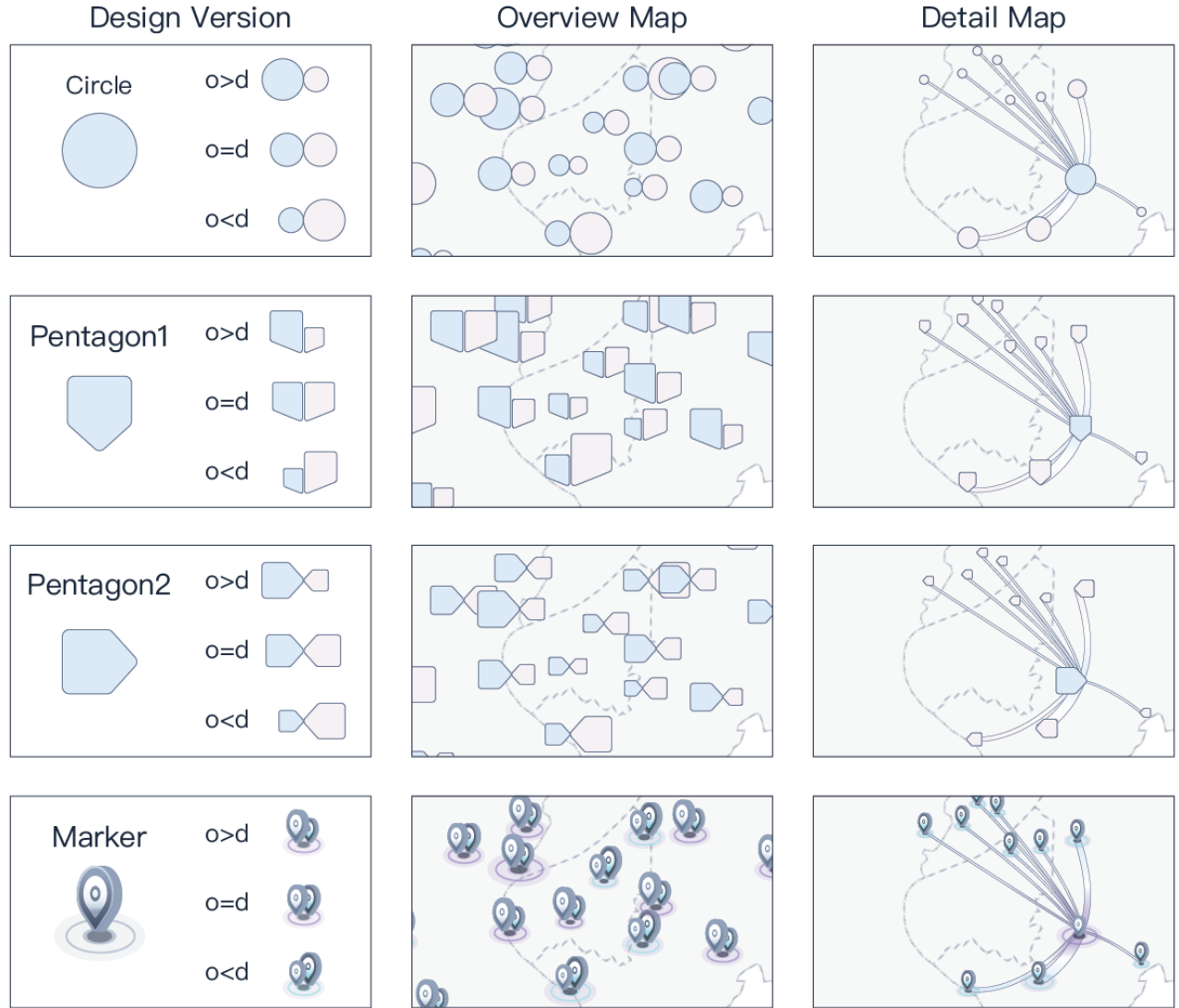


Fig. 4. The 2D Failures. The left column represents the iterative prototypes evaluated: (A) 2D Circles, (B-C) Mirrored Geometric Glyphs (2D Pentagon 1 & 2), and (D) Pseudo-3D Markers. The middle and right illustrate the encoding logic for coincident flows (O_iD , $O=D$, O_iD) and their performance in dense overview maps versus sparse detail maps. Observe that the 2D approaches (A-C) experience occlusion in the overview state.

Furthermore, the reliance on area encoding proved perceptually inefficient. Extensive empirical evaluations by Saket et al. [19] confirm that users judge area size with far less accuracy than length or position. Compounded by the computational cost of real-time symbol recalculation, the display became both unstable and imprecise. These failures led to the rejection of dynamic 2D glyphs in favor of static alternatives.

B. 2D Mirrored Geometric Glyphs (Pentagon 1 & 2)

To address the instability of the previous iteration, mirrored pentagonal glyphs were anchored directly to AOI centroids. While spatially consistent, this design introduced severe perceptual bottlenecks. The primary failure lay in the use of

irregular area encoding. As Heer and Bostock [20] demonstrate, area channels, particularly for non-standard shapes, result in significantly higher error rates than position or length.

The mirrored configuration exacerbated this by requiring mental transformation. Gleicher [21] notes that separating comparative elements, even symmetrically, imposes high cognitive overhead that degrades judgement accuracy. Confronted with this difficulty, users reverted to simplified heuristics, often confusing shape height for total volume—a cognitive bias documented by Padilla et al. [22]. Combined with extensive visual clutter at the city scale, these limitations rendered the glyphs ineffective for professional analysis.

C. Pseudo-3D Markers with Categorical Encoding

The third design introduced pseudo-3D markers rendered utilizing categorical encoding. It has a central elliptical pattern that features a central elliptical pattern mimicking concentric water ripples. The approach abandoned proportional scaling and reduced the encoding to three categories: $O < D$, $O = D$, and $O > D$. Small changes in size and a colored base ripple indicated the dominant flow.

This reduction caused severe information loss; a 1% difference became visually indistinguishable from a 99% gap, which prevented any meaningful quantitative reading. Users could not compare actual values within an AOI or across AOIs because the icons appeared almost identical.

The design also failed spatially. At high zoom levels, the ripple was centered between markers rather than aligned with its variable. This violation of proximity caused users to dismiss the data cue as a rendering artifact, a failure of interpretability consistent with Padilla et al. [22]. At lower zoom levels, the cue's transparency and reduced surface area rendered it effectively invisible. This validates Szafir's [23] findings on the perceptual limits of visualization, which confirm that small-field color encodings are insufficient for reliable discrimination.

Furthermore, the markers continued to overlap significantly, resulting in high rates of selection errors during data drill-down. While this design eliminated positional drift, it sacrificed essential quantitative fidelity, demonstrating that pseudo-3D glyphs cannot effectively resolve the comparison of coincident values.

D. 3D Pillars

The inherent limitation failures of 2D designs necessitated a departure from the planar map. Following established precedents in space-time visualization [24], this study adopted vertical extrusion as the only scalable method to disentangle coincident attributes without distorting their spatial topology.

The resulting "Pillars" design relies on three design principles:

- **Linear Encoding:** Magnitude is mapped to pillar height rather than volume. Validating this choice, recent empirical studies [7] confirm that 3D bars support significantly higher accuracy for quantitative comparison than volumetric encodings, justifying the rejection of cone-based geometries.
- **Spatial Stability:** Pillars are separated by a fixed, immutable offset. Unlike the dynamic shifting of earlier prototypes, this adherence to spatial constancy minimizes cognitive load and preserves the user's mental map during navigation [18].
- **Constrained Viewpoint:** The system employs a fixed 60° pitch. This exposes height differences that collapse in top-down views while avoiding the disorientation typical of free-flight navigation [25]. Collectively, these constraints mitigate occlusion, allowing for the direct, stable comparison of coincident values.

E. 3D Cones

A second 3D option encoded magnitude using the volume of a cone. Despite its visual appeal, this design relied on a perceptual channel widely recognized as inefficient. The cone geometry created interference between interdependent visual

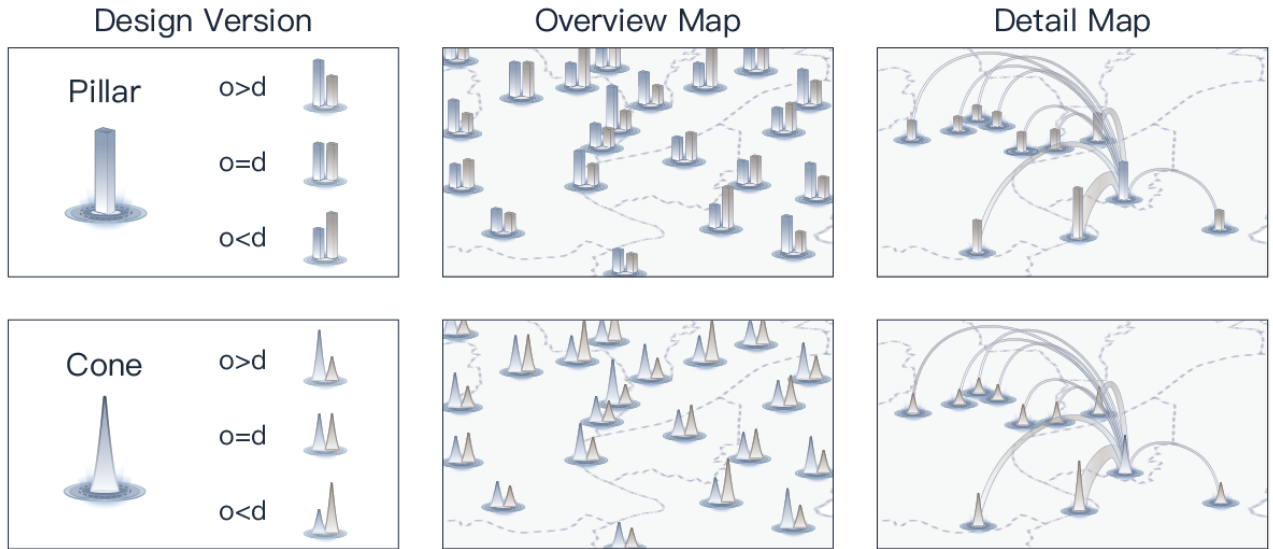


Fig. 5. The 3D Solution & Comparison. The left column represents the iterative prototypes evaluated: (E) the proposed 3D pillars and (F) 3D cones. The cone geometry (F) introduces visual ambiguity when comparing precise heights to the linear extrusion of pillars (E).

dimensions. Because magnitude changes altered both the slope and the base-to-apex relationship, the design lacked a stable baseline for comparison. This "floating" geometry forced users to perform complex shape integration rather than simple height estimation. Consequently, participants consistently struggled to distinguish values. Volumetric cones were therefore rejected in favor of the pillar-based encoding. This decision was not aesthetic but functional: it aligns with empirical findings that length-based representations, even in 3D environments, provide the perceptual precision necessary for reliable quantitative analysis [7].

V. INTERACTION

The Pillars encoding separates origin and destination values at each node, but visual complexity re-emerges once flows are drawn. A full N:N network of 3D arcs produces heavy overlap and depth ambiguity, making static clutter-reduction techniques insufficient. Interaction therefore becomes central to the design: it must control when and how flow structures appear so that users can interpret them without navigating a wall of overlapping geometry.

In preliminary tests, transparency (often termed 'X-ray shading'), which is a passive occlusion-management strategy used in 3D visualization, performed poorly. When hundreds of arcs occupy the same volume, reducing opacity simply adds visual noise and makes isolating individual paths more difficult [26].

To address this, the interaction model was designed around progressive disclosure: prioritizing scene stability while utilizing interaction to progressively narrow the analytic space from many-to-many to one-to-many. Two complementary mechanisms support this approach: coordinated views and a focus-plus-context drill-down.

A. Coordinated Views

The system links the 3D map with an adaptive 2D view, allowing each view to address the perceptual task it handles best [27]. The map preserves geography and exposes spatial

structure. The diagram handles ranked comparisons and proportional judgement, which are difficult to perform using 3D geometry alone due to perspective distortion.

In the Overview State, the 2D panel presents a horizontal bar chart ranking the 193 AOIs' OD flows separately. This prioritizes the rapid identification of dominant districts, a task where aligned linear scales outperform 3D geometry. Upon selecting an AOI, the system transitions to the Detail State. The 2D view transforms into a Sankey diagram, while the map generates the corresponding 3D flow arcs.

The views communicate bidirectionally. Selecting a pillar in the 3D map updates the 2D view to show destination flows for that region. Conversely, selecting a flow path (link) in the Sankey diagram triggers a global highlight on the map, diminishing non-selected elements to background opacity. This allows users to inspect specific flow details, such as precise commuter counts, without losing the overall spatial context.

B. Focus+Context Drill-Down

The drill-down mechanism is designed to manage the complexity of dense urban networks. Upon selecting an origin 3D pillar or 2D bar, the system triggers a state transition (tracked via breadcrumb navigation, e.g., Hangzhou $\rightarrow$ West Lake District) that clarifies the surrounding flows.

From Many-to-Many (N:N) to One-to-Many (1:N) N:N to 1:N: In the focused state, only the selected origin and its connected destinations remain visually strong; opacity for unrelated elements is reduced. This shift effectively reduces the problem from a cluttered N:N network to a comprehensible 1:N pattern. This interactive filtering strategy aligns with established practices in graph visualization, which suggest that reducing edge density on demand is far more effective than static layout algorithms alone [11]. This removes uncertainty about which nodes are involved before the user engages with the detailed flow arcs.

Once the node structure is established, 3D arcs link the origin to its destinations. The width of each arc encodes flow magnitude, and lifting these paths into the Z-dimension facilitates separation [11]. To manage visual clutter and ensure

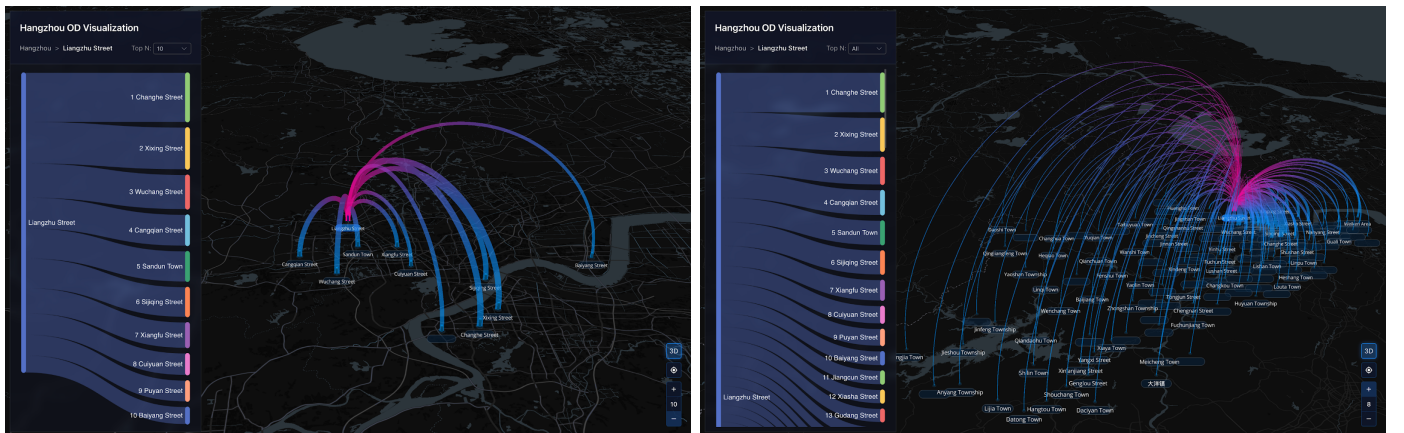


Fig. 6. Figure 6: "Top 10" vs. "All" Comparison.

scalability, the system incorporates a dynamic density filter (Top-X Control). This feature allows users to toggle between the top 10, 30, 50, or 100 flows based on volume. This capability enables the interface to support both high-level trend analysis (showing only major arteries) and granular inspection (revealing “capillary” movements) without inducing computational lag or visual overload.

While 3D extrusion resolves the topological conflict of coincident nodes, unconstrained 3D navigation often imposes a heavy cognitive load and leads to spatial disorientation. To avoid this, the system uses a constrained navigation model with a fixed 60° pitch. Users can pan and zoom but cannot freely orbit the scene. This constraint preserves a stable viewpoint for height comparison while still enabling depth perception. As the map moves, motion parallax provides kinetic depth cues that help users distinguish overlapping pillars and reveal partially occluded arcs without requiring manual rotation [28].

To support precise spatial localization, the interface also offers a 2D/3D perspective toggle. The 3D mode emphasizes magnitude assessment and flow separation, whereas the 2D mode collapses the view into an orthographic projection that restores exact geographic footprints. These complementary modes align with recent findings in immersive analytics, which show that alternating between exocentric (2D) and perspective (3D) views improves tasks requiring both structural understanding and fine-grained spatial lookup. By combining constrained 3D interaction with an immediate 2D fallback, the system balances quantitative clarity with geographic precision [29].

VI. IMPLEMENTATION AND EVALUATION

The system architecture adopts a hybrid rendering strategy to balance geospatial precision with high-performance visualization. The application utilizes MapLibre GL JS for the basemap and static geometry (pillars) and deck.gl for high-frequency dynamic overlays (Flow Arcs). This separation of concerns allows us to leverage the specific strengths of each library: MapLibre’s efficient tile-based vector rendering and deck.gl’s WebGL-powered instancing.

A. Texture-Atlas Optimization for Pillars

Rendering the pillars with a vertical comparison gradient was a key design requirement for facilitating quantitative comparison. Although deck.gl’s ColumnLayer is optimized for using instancing to render solid-colored geometry, applying a vertical gradient usually necessitates injecting custom WebGL shaders, which introduces significant engineering complexity. To maintain a streamlined codebase without compromising visual fidelity, the system leverages MapLibre’s fill-extrusion-pattern property. The HTML5 Canvas API was used to programmatically create a discretized 256x256 gradient ramp, which was then registered as a sprite using map.addImage. The engine can batch-render all 386 pillar geometries in a single draw call thanks to this efficient use of MapLibre’s internal Texture Atlas. Using standard API features instead of custom

shader modules, this method effectively produced the desired aesthetic—a clear vertical linearity comparison.

B. Instanced Rendering for Flow Arcs

Interactive drill-downs require rendering hundreds of 3D arcs in real time. The system added deck.gl’s ArcLayer because standard vector tile libraries don’t support high-fidelity 3D splines. This layer utilizes WebGL Geometry Instancing, enabling the system to render thousands of arc segments while uploading the geometry buffer to the GPU only once. During the “Focus+Context” transitions, the system updates only the instance attributes (source/target coordinates, colors) rather than reconstructing the DOM or geometry, ensuring instant visual feedback during high-frequency interactions.

Rendering performance was evaluated on standard commodity hardware (e.g., MacBook Pro M1, Windows 10 with integrated graphics) running Chrome 114+.

- **Rendering Stability:** In the “Overview” state (all 386 pillars visible), the application maintains a stable 60 FPS. The texture-atlas optimization ensures that the basemap thread remains unblocked during pan and zoom operations.
- **Interaction Latency:** In the “Detail” state, highlighting specific flows involves filtering and re-rendering arc instances. Performance profiling indicates that frame rates remain above 30 FPS even when rendering up to 500 simultaneous arcs.
- **Scalability:** The hybrid architecture successfully decouples the static semantic layer (AOIs) from the dynamic flow layer. This confirms that the approach is theoretically scalable to larger datasets, limited only by the client-side memory required to store the raw topology graph rather than GPU rendering bottlenecks.

VII. DISCUSSION

A. Generalizability of the Framework

While this study relies on urban mobility data, the “Pillars” design pattern addresses a fundamental structural challenge applicable to any domain involving spatially aggregated interaction networks.

As demonstrated by our processing of 400,000 raw trajectories into 193 semantic units, raw movement data is often unintelligible until it is aggregated into meaningful spatial entities. The Pillars framework excels precisely in these scenarios where massive mobility records must be consolidated into fixed nodes that act simultaneously as sources and sinks.

This topological conflict—where multiple opposing attributes (e.g., In vs. Out) occupy the same geographic coordinate—is common in various fields:

- **International Trade:** Ports or nations can be visualized to simultaneously compare import vs. export volumes (trade balance) at a single location, resolving the ambiguity of “entrepôt” trade.
- **Demography and Migration:** Cities or census tracts serve as hubs for human mobility. The framework could clarify immigration vs. emigration flows, helping demographers

assess net migration rates without losing the context of total volume.

- **Financial Geography:** Regional banking nodes could visualize capital inflow vs. outflow, allowing for the rapid identification of liquidity centers versus sinks.

By abstracting the problem as the “comparative visualization of coincident aggregated attributes”, our solution extends beyond traffic analysis, offering a generalized method for exploring, comparing, and drilling down into any dense, bidirectional geospatial network.

Future work will focus on extending the framework to handle dynamic time-series data and integrating “small multiple” views for the Sankey component to reduce clutter during multi-node selection. However, in its current form, the Pillars framework offers a powerful, scalable pattern for the growing field of urban mobility analytics.

B. Limitations

This study acknowledges several limitations in the current study. Primary among these is the evaluation methodology. While the design decisions were grounded in established perceptual theories (e.g., Cleveland & McGill’s elementary perceptual tasks), the validation relied on expert critique and internal stakeholder review rather than a controlled quantitative user study. Future work should involve formal laboratory experiments to measure error rates and response times when comparing “Pillars” against other 3D encodings. Second, the scalability of the visualization for extremely dense networks remains a challenge. While the “Top-X” filtering addresses rendering performance, severe visual clutter may still occur in “All-Flows” mode when the number of coincident nodes exceeds a certain threshold. Finally, the current interaction model focuses on desktop environments; adapting these 3D interactions for mobile or touch-based interfaces requires further investigation.

C. Note on Technical Availability and Reproducibility

To validate the real-time interaction capabilities and the fluidity of the Focus+Context transitions described in this report, an interactive web demonstration utilizing synthetic data (mimicking real-world density) is publicly available at <https://od-visualization-beta.vercel.app/>. Furthermore, to ensure the reproducibility of the rendering techniques, the complete source code—including the gradient texture generation and deck.gl overlay management—is archived at <https://github.com/yutonghan19951123/od-visualization>.

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